

**Model run note:** This version was last run on **29/06/2026** with **TUFLOW FV build 2025.2.1**.

- **Control file used to run the model:**

`.../Complete_Model/TUFLOWFV/runs/HYD_aed.fvc`

- **Run log file (more information):**

`.../Complete_Model/TUFLOWFV/runs/log/HYD_aed.log`

## Tutorial: Running TUFLOW FV with the AED Water-Quality Module

This notebook extends the official **BMT TFV tutorial** ([tfv.readthedocs.io](https://tfv.readthedocs.io)) by enabling the **AED (Aquatic EcoDynamics)** water-quality module on top of the standard TUFLOW FV estuary test case. The hydrodynamic model uses the **default configuration**; AED runs with **standard settings**. The aim is a minimal, end-to-end workflow to confirm the coupled hydro–WQ model is configured correctly and to visualise the outputs.

### This notebook covers

1. Loading TUFLOW FV + AED NetCDF results ( `HYD_aed.nc` ).
2. Verifying the model ran by inspecting the mesh and a depth-averaged WQ snapshot (dissolved oxygen).
3. Diagnostic plots for **water-quality state variables** — Oxygen, TN, GPP, TSS, TP, TCHLA, Turbidity, PAR, Light — alongside velocity and density.
4. Use this as a **starting point** to customise TUFLOW FV + AED runs for your own estuarine application.

**Source files referenced:** `HYD_aed.nc` (results) and `polyline.csv` (curtain transect).

## What you will learn

1. **Load** a TUFLOW FV NetCDF result file as an xarray Dataset.
2. **Map snapshots** - depth-averaged surface plots (example: dissolved oxygen).
3. **Surface vs. seabed vs. depth-averaged** - 3×3 grid contrasting Velocity, TSS and TP.
4. **Timeseries** - extract WQ state variables ( `WQ_PHY_MIXED` , `WQ_DIAG_TOT_TSS` , `WQ_DIAG_PHY_TCHLA` ) at named stations.
5. **Sheets & statistics** - depth-averaged GPP sheets and 99th-percentile maps over a time window.
6. **Curtains** - vertical slices of WQ fields along a user-defined polyline transect.
7. **Profile + timeseries** - vertical TSS structure at one station, plus its depth-averaged history.
8. **Hovmoller** - depth vs. time diagrams of TSS and Turbidity at a station.
9. **Animations** - export an MP4 of the evolving Turbidity field.
10. **Interactive views** - Jupyter-widget plot with a time slider.

# Requirements

```
pip install tfv xarray matplotlib seaborn netCDF4 ipyml
# For MP4 export:
sudo apt install ffmpeg
```

## How to use

1. Open in **JupyterLab** or **VS Code** with the Jupyter extension.
2. Update the `model_folder` path in Section 1 to point at your local results directory.
3. Run **Run All** the first time, then experiment by swapping the WQ variable, station, time, or colour ranges in each cell.

**Note for GitHub viewers:** Outputs are committed with the notebook so the rendered figures display on GitHub. Re-run after pulling to refresh against new model results.

## 1. Load the model results

We open the NetCDF file directly with `xarray`. The `decode_times=False` flag is important because TUFLOW FV writes time in a non-CF convention — the `tfv` accessor decodes it on the next step. The displayed `Dataset` lists all available state variables (hydrodynamic + AED water-quality) and their dimensions.

```
getfattr: /media/fileshare/AED/RONY/15_TUFLOWFV_aed/15_TUFLOWFV/Tutorial_Module_05_Estuary_Module/Tutorial_Module_05_Estuary_Module/Complete_Model/TUFLOWFV/results/HYD_aed.nc: Operation not supported
```

xarray.Dataset

► **Dimensions:** (Time: 145, NumCells2D: 1375, MaxNumCellVert: 4, NumCells3D: 6453, NumVert2D: 1419, NumLayerFaces3D: 7828)

► **Data variables:** (139)

▼ **Attributes:**

|             |                              |
|-------------|------------------------------|
| Origin :    | Created by TUFLOWFV          |
| Type :      | Cell-centred TUFLOWFV output |
| spherical : | true                         |
| Dry depth : | 0.01                         |

## Attach the `tfv` accessor

Calling `ds.tfv` (or wrapping the dataset in `tfv.xarray.TfvDomain`) decodes the model grid, time axis, and 3D layer faces. The returned `TfvDomain` is what we use for every plot below.

xarray.Dataset

► Dimensions: (Time: 145, NumLayerFaces3D: 7828, NumCells2D: 1375, NumCells3D: 6453)

▼ Coordinates:

**Time** (Time) datetime64[us] 2011-05-01 ... 2011-05-07



► Data variables: (127)

▼ Attributes:

Origin : Created by TUFLOWFV  
 Type : Cell-centred TUFLOWFV output  
 spherical : true  
 Dry depth : 0.01

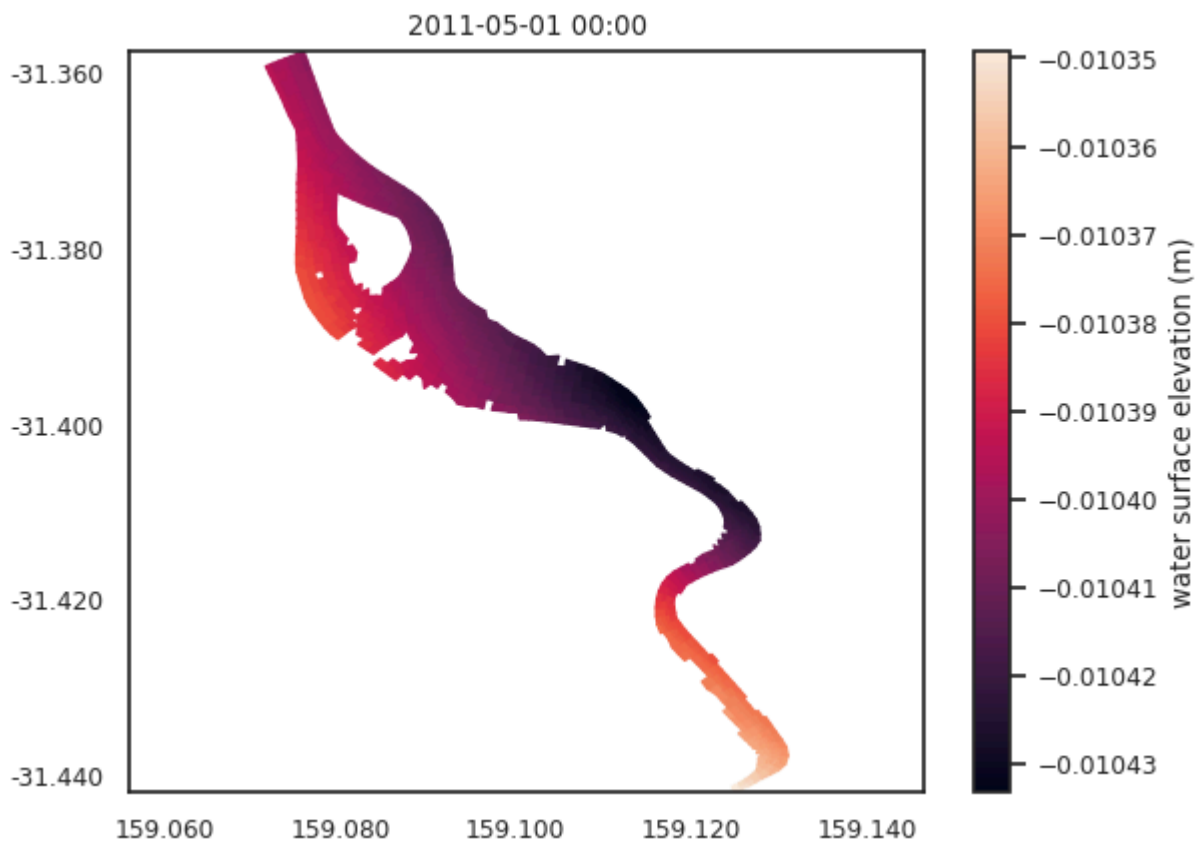
◀ TUFLOW FV domain xarray accessor object ▶

## 2. Quick map snapshots

`fv.plot()` with no arguments renders the model mesh, which is a good sanity check that the grid loaded correctly. With a variable name + time it plots a depth-averaged horizontal map — below we use **dissolved oxygen** (`WQ_OXY_OXY`) at the model surface as a first WQ snapshot.

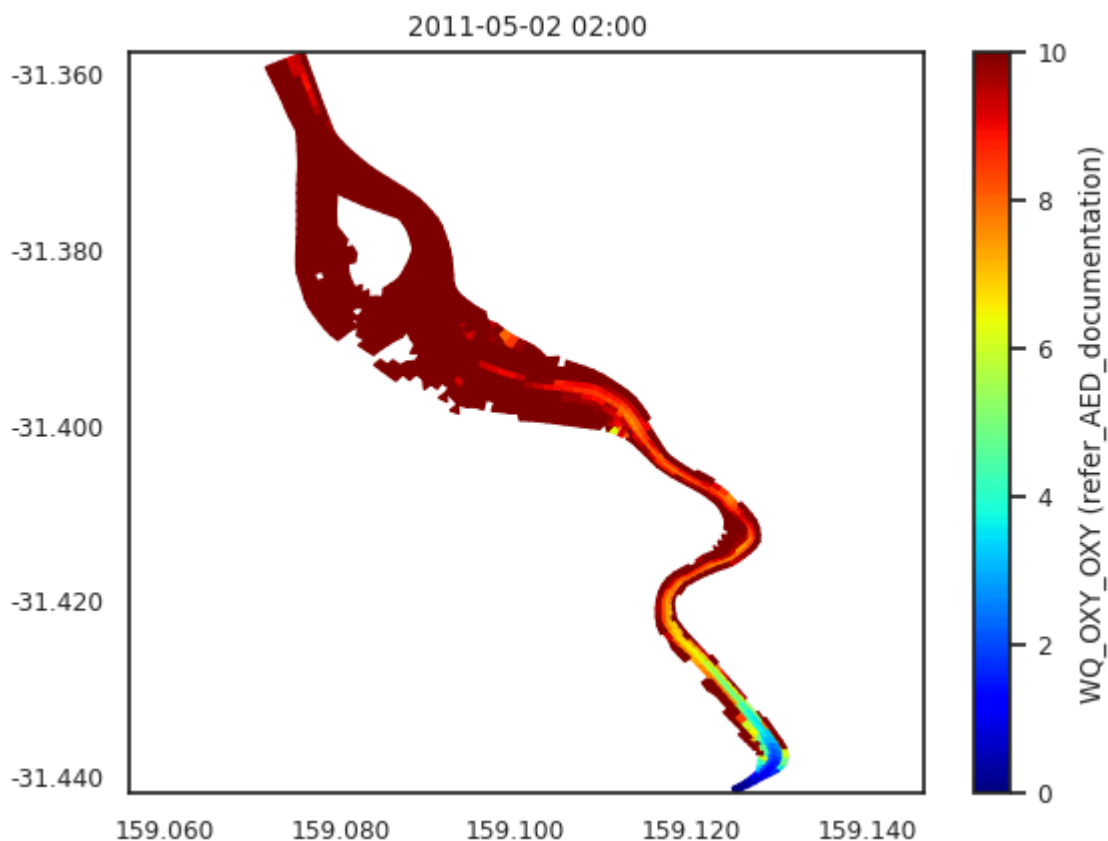
<tfv.visual.SheetPatch at 0x7988d7f6e750>

Ignoring fixed x limits to fulfill fixed data aspect with adjustable data limits.



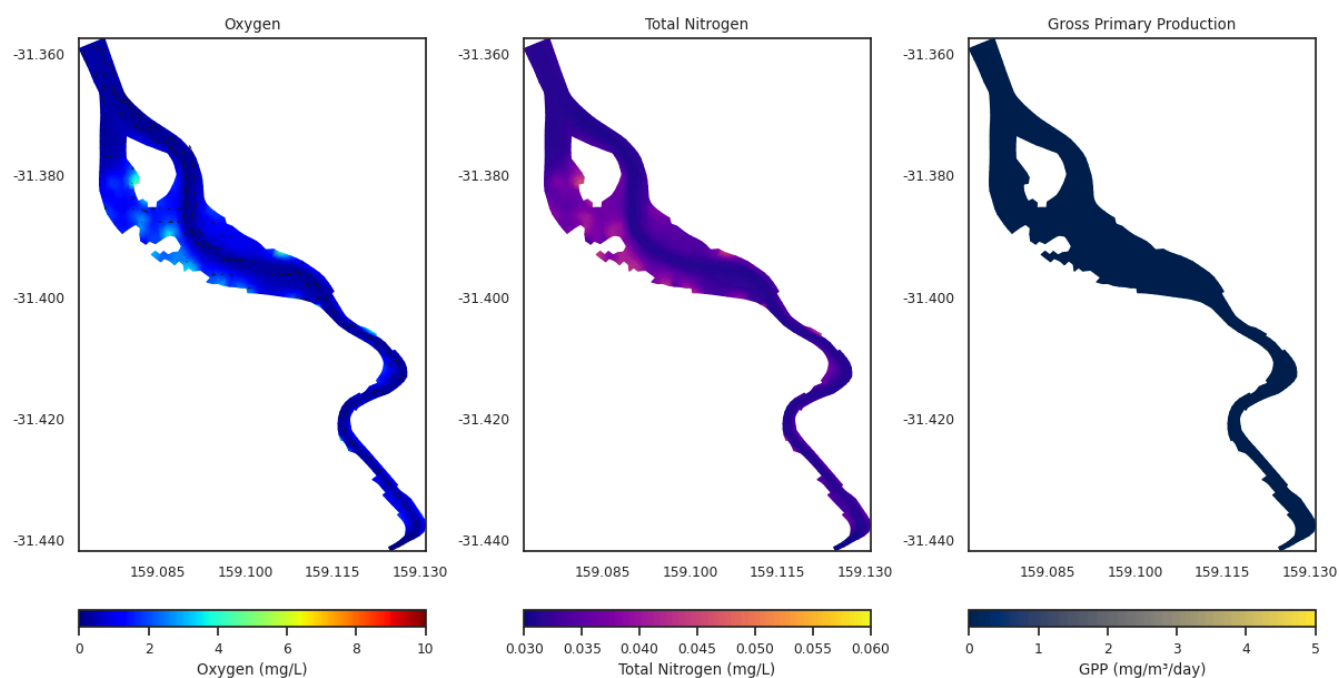
<tfv.visual.SheetPatch at 0x7988d7f6db20>

Ignoring fixed x limits to fulfill fixed data aspect with adjustable data limits.



### Three-panel side-by-side view

A common pattern is to compare three related variables at the same time. Here we show **Oxygen** (with velocity vectors overlaid), **Total Nitrogen**, and **Gross Primary Production (GPP)**. We use `datum='depth'` with `limits=(0, 1)` to render the **surface 1 m** (try `datum='height'` to flip to the bottom 1 m). `plot_vector` overlays velocity arrows on top of the oxygen field.



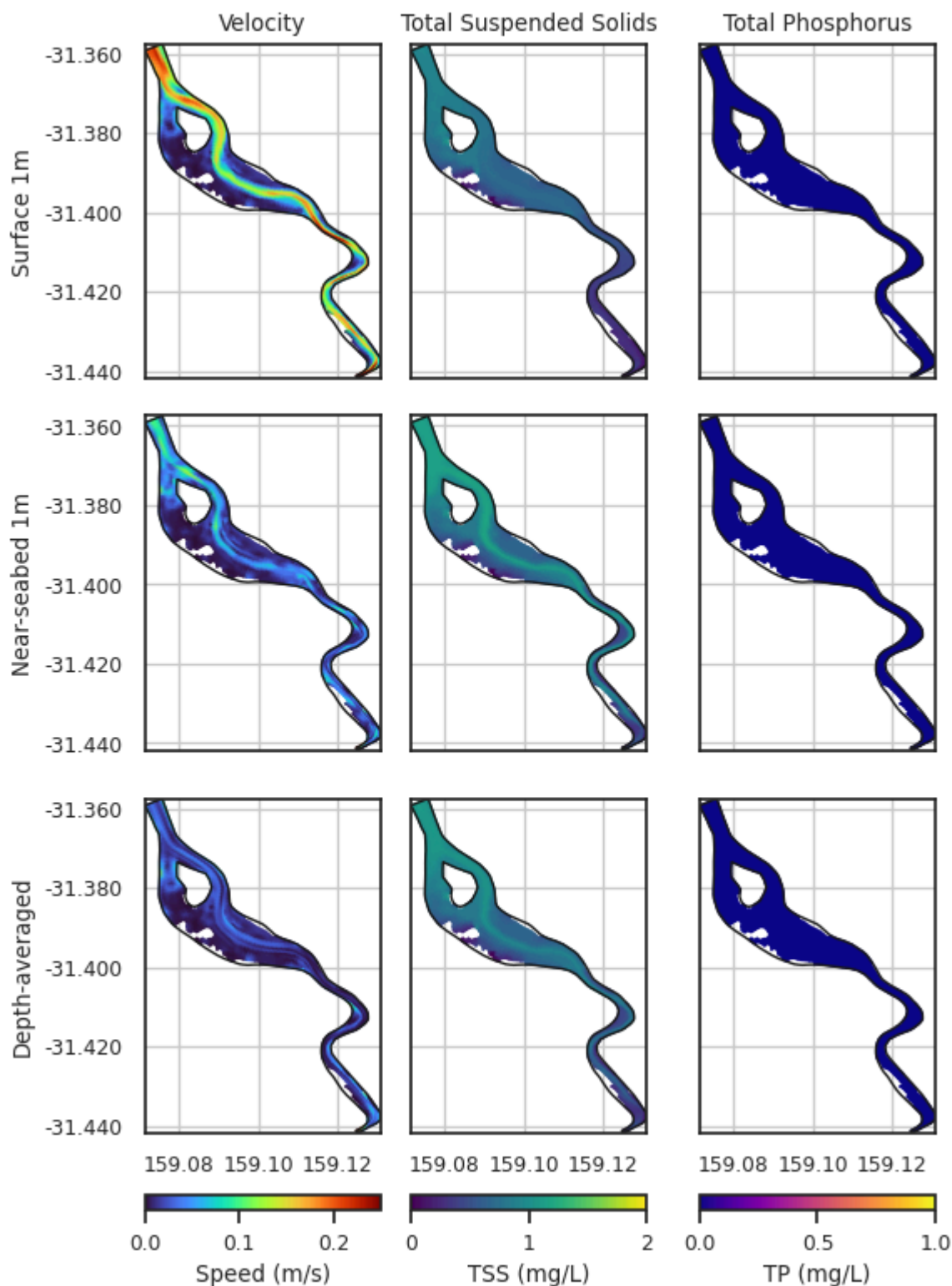
### 3. Surface vs. seabed vs. depth-averaged

A 3×3 grid contrasting three variables at three depth treatments:

- **Columns** - Velocity ( `V` ), Total Suspended Solids ( `WQ_DIAG_TOT_TSS` ), Total Phosphorus ( `WQ_DIAG_TOT_TP` ).
- **Top row** - surface 1 m ( `datum='depth'` , `limits=(0, 1)` ).
- **Middle row** - near-seabed 1 m ( `datum='height'` , `limits=(0, 1)` ).
- **Bottom row** - depth-averaged (default).

This is a quick **stratification-vs-mixing diagnostic for water-quality fields** — vertical differences in any column reveal whether TSS/TP are uniformly mixed or settling/accumulating at depth.

### Example TUFLOW FV water quality result



# 4. Point timeseries extraction

`fv.get_timeseries()` interpolates the chosen variables to one or more (x, y) stations across the full simulation time. Here we extract three WQ state variables — phytoplankton biomass ( `WQ_PHY_MIXED` ), total suspended solids ( `WQ_DIAG_TOT_TSS` ), and chlorophyll-a ( `WQ_DIAG_PHY_TCHLA` ) — at stations `P1` and `P2` . The result is an `xarray.Dataset` indexed by ( `Time`, `Location` ) — easy to convert to a pandas DataFrame for tabular plotting or CSV export.

```
Extracting timeseries, please wait:  0%|          | 0/145 [00:00<?, ?it/s]Extracting timeser
ies, please wait: 100%|██████████| 145/145 [00:04<00:00, 35.74it/s]
```

`xarray.Dataset`

► Dimensions: (Time: 145, Location: 2)

▼ Coordinates:

|          |            |                |                           |  |  |
|----------|------------|----------------|---------------------------|--|--|
| Time     | (Time)     | datetime64[us] | 2011-05-01 ... 2011-05-07 |  |  |
| Location | (Location) | <U2            | 'P1' 'P2'                 |  |  |
| x        | (Location) | float64        | 159.1 159.1               |  |  |
| y        | (Location) | float64        | -31.39 -31.39             |  |  |
| z        | (Location) | float32        | -1.068 -4.886             |  |  |

▼ Data variables:

|                  |                  |         |                                     |  |  |
|------------------|------------------|---------|-------------------------------------|--|--|
| WQ_PHY_MIXED     | (Time, Location) | float64 | 0.0 0.0 0.2 0.2 ... 0.2 0.2 0.2 0.2 |  |  |
| WQ_DIAG_TOT_T... | (Time, Location) | float64 | 0.004 0.004 ... 0.7959 0.9622       |  |  |
| WQ_DIAG_PHY_...  | (Time, Location) | float64 | 0.048 0.048 0.048 ... 0.048 0.048   |  |  |

▼ Attributes:

Origin :

Type :

spherical :

Dry depth :

Datum :

Limits :

Agg Fn :

Timeseries extracted from TUFLOWFV cell-centered output using `tfv` python tools

Timeseries cell from TUFLOWFV Output

true

0.01

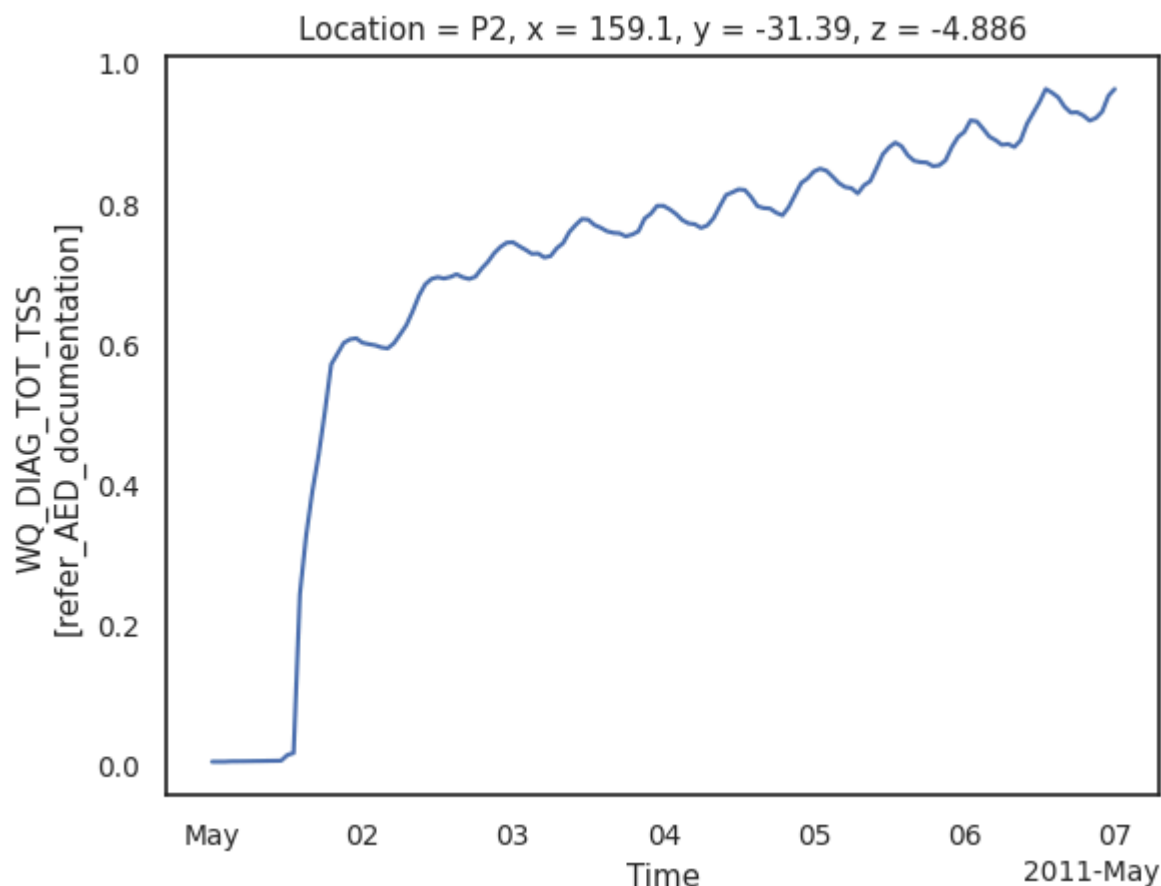
sigma

(0, 1)

mean

A quick sanity-check plot — TSS timeseries at station P2 using `xarray`'s native plotting.

```
[<matplotlib.lines.Line2D at 0x7988c85cc7d0>]
```



## 5. Sheets and statistics

A **sheet** is the depth-averaged (or depth-limited) horizontal slab of a variable, returned as a 2-D field per timestep — handy for animation frames or aggregation. Here we build a GPP ( `WQ_DIAG_PHY_GPP` ) sheet for the first two model days.

**Statistics** ( `min` , `mean` , `max` , percentiles like `p99` ) collapse the time axis to produce a single map showing extreme-value behaviour — useful for identifying productivity hotspots or stress events. Below we run them on near-seabed GPP and plot the 99th-percentile field.

xarray.Dataset

► Dimensions: (Time: 72, NumLayerFaces3D: 7828, NumCells2D: 1375)

▼ Coordinates:

|             |        |                |                               |                                                                                                                                                                         |
|-------------|--------|----------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Time</b> | (Time) | datetime64[us] | 2011-05-02 ... 2011-05-04T... |   |
|-------------|--------|----------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

▼ Data variables:

|                 |                         |         |                                     |                                                                                                                                                                         |
|-----------------|-------------------------|---------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ResTime         | (Time)                  | float64 | 1.87e+05 1.87e+05 ... 1.871...      |   |
| layerface_Z     | (Time, NumLayerFaces3D) | float32 | ...                                 |   |
| stat            | (Time, NumCells2D)      | int32   | ...                                 |   |
| WQ_DIAG_PHY_... | (Time, NumCells2D)      | float64 | 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 ... |   |

▼ Attributes:

Origin : Created by TUFLOWFV  
 Type : Cell-centred TUFLOWFV output  
 spherical : true  
 Dry depth : 0.01

◀  ▶  
 TUFLOW FV domain xarray accessor object

...extracting sheet data: 100%|  | 145/145 [00:01<00:00, 111.77it/s]

xarray.Dataset

► Dimensions: (Time: 1, NumLayerFaces3D: 7828, NumCells2D: 1375)

▼ Coordinates:

|             |        |                |            |                                                                                                                                                                             |
|-------------|--------|----------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Time</b> | (Time) | datetime64[us] | 2011-05-01 |   |
|-------------|--------|----------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

▼ Data variables:

|                 |                         |         |                                     |                                                                                                                                                                             |
|-----------------|-------------------------|---------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ResTime         | (Time)                  | float64 | 1.87e+05                            |   |
| layerface_Z     | (Time, NumLayerFaces3D) | float32 | ...                                 |   |
| stat            | (Time, NumCells2D)      | int32   | ...                                 |   |
| WQ_DIAG_PHY_... | (Time, NumCells2D)      | float64 | 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 ... |   |
| WQ_DIAG_PHY_... | (Time, NumCells2D)      | float64 | 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 ... |   |
| WQ_DIAG_PHY_... | (Time, NumCells2D)      | float64 | 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 ... |   |
| WQ_DIAG_PHY_... | (Time, NumCells2D)      | float64 | 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 ... |   |
| WQ_DIAG_PHY_... | (Time, NumCells2D)      | float64 | 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 ... |   |
| WQ_DIAG_PHY_... | (Time, NumCells2D)      | float64 | 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 ... |   |
| WQ_DIAG_PHY_... | (Time, NumCells2D)      | float64 | 0.0 0.0 0.0 0.0 ... 0.0 0.0 0.0 ... |   |

▼ Attributes:

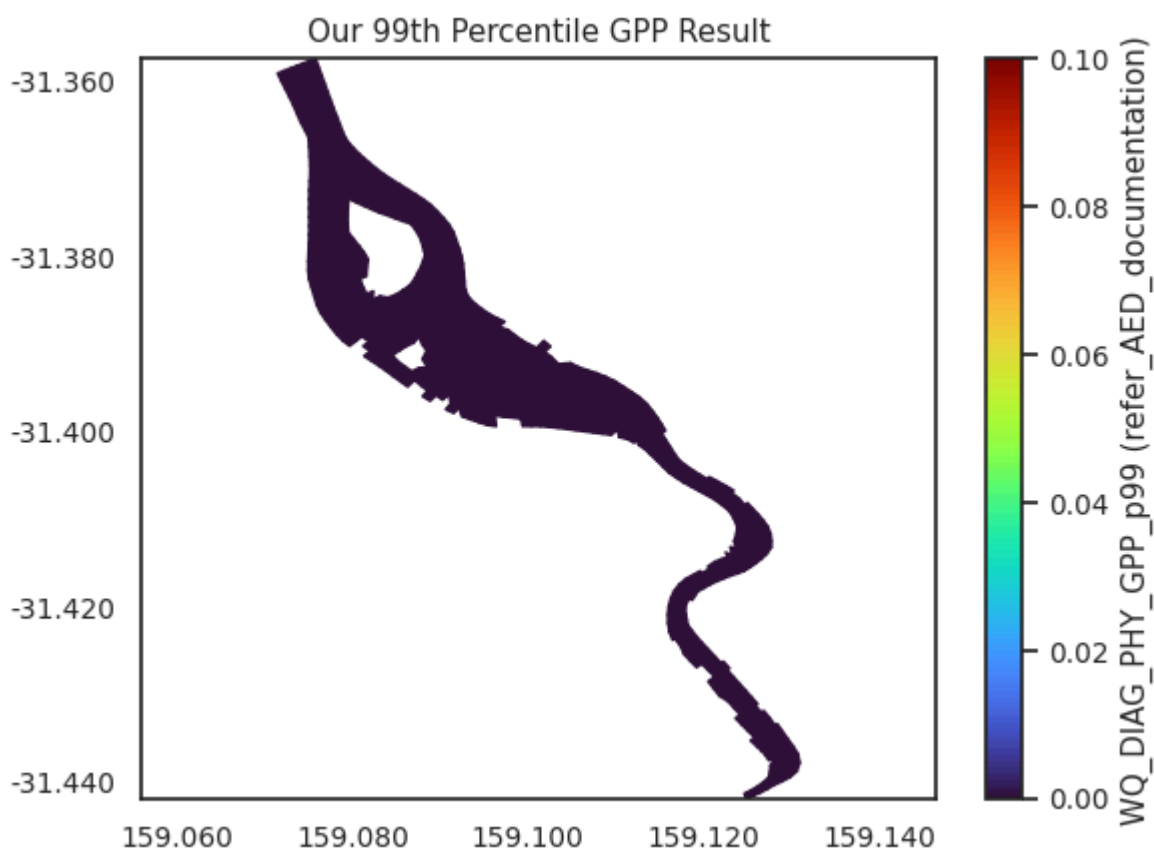
Origin : Created by TUFLOWFV  
 Type : Cell-centred TUFLOWFV output  
 spherical : true  
 Dry depth : 0.01

◀  ▶  
 TUFLOW FV domain xarray accessor object



```
Text(0.5, 1.0, 'Our 99th Percentile GPP Result')
```

Ignoring fixed x limits to fulfill fixed data aspect with adjustable data limits.

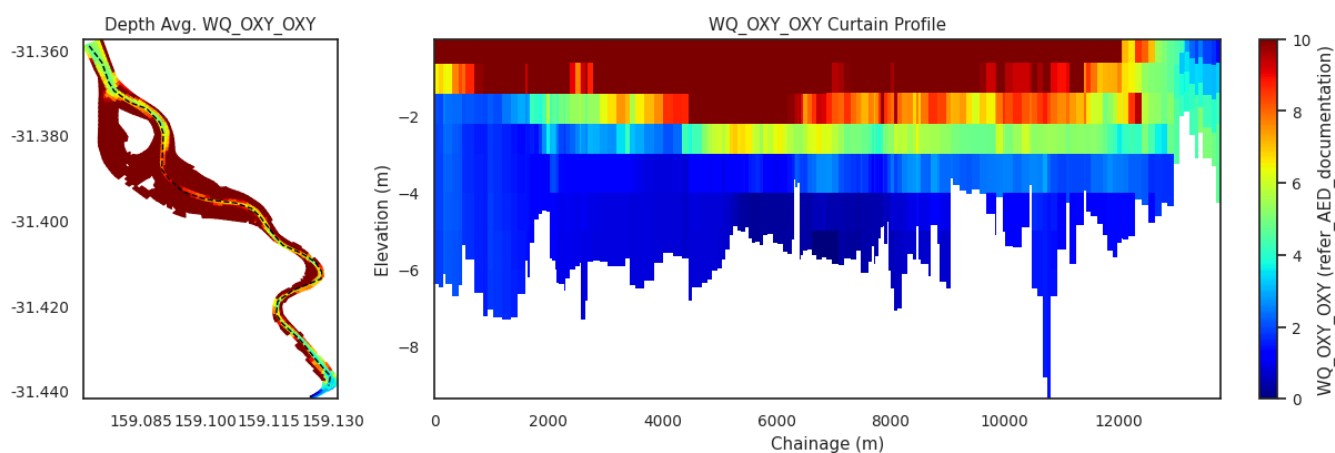


## 6. Curtain plots (vertical slices)

A **curtain** is a vertical cross-section along an arbitrary polyline (loaded here from `polyline.csv` — two columns of `x`, `y` coordinates). It exposes the stratification structure that map views collapse away. The seaborn theme below also normalises font sizes for publication-quality figures.

### Map + curtain side-by-side

Pairing the depth-averaged map (left, narrow) with the curtain (right, wide) makes it obvious *where* the water-quality structure you're seeing happens. The black dashed line on the map is the polyline used for the curtain. Below we use **dissolved oxygen** (`WQ_OXY_OXY`) to spot any near-bed hypoxic pockets along the transect.



## 7. Profile and timeseries at a station

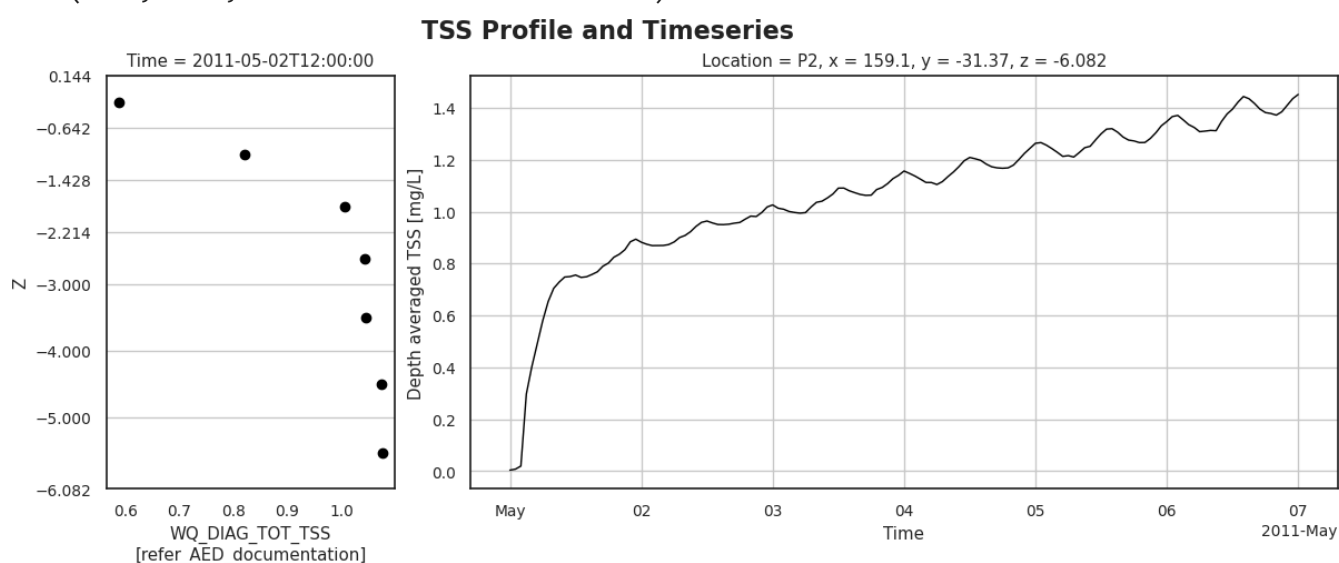
For a deeper look at one location we combine:

- a **vertical profile** ( `fv.get_profile` ) — TSS vs. depth at a single time, with model layer faces marked on the y-axis to show vertical resolution.
- a **depth-averaged timeseries** at the same station, providing temporal context.

Below we extend the location dictionary to a chain of stations along the estuary ( `P1 ... P8` ) so the same workflow can be reused at any point. The variable shown is `WQ_DIAG_TOT_TSS` — swap `var` to any other WQ field for a similar diagnostic.

```
Extracting timeseries, please wait: 0%|          | 0/145 [00:00<?, ?it/s]Extracting timeseries, please wait: 100%|██████████| 145/145 [00:01<00:00, 104.13it/s]
```

```
Text(0.45, 0.98, 'TSS Profile and Timeseries')
```

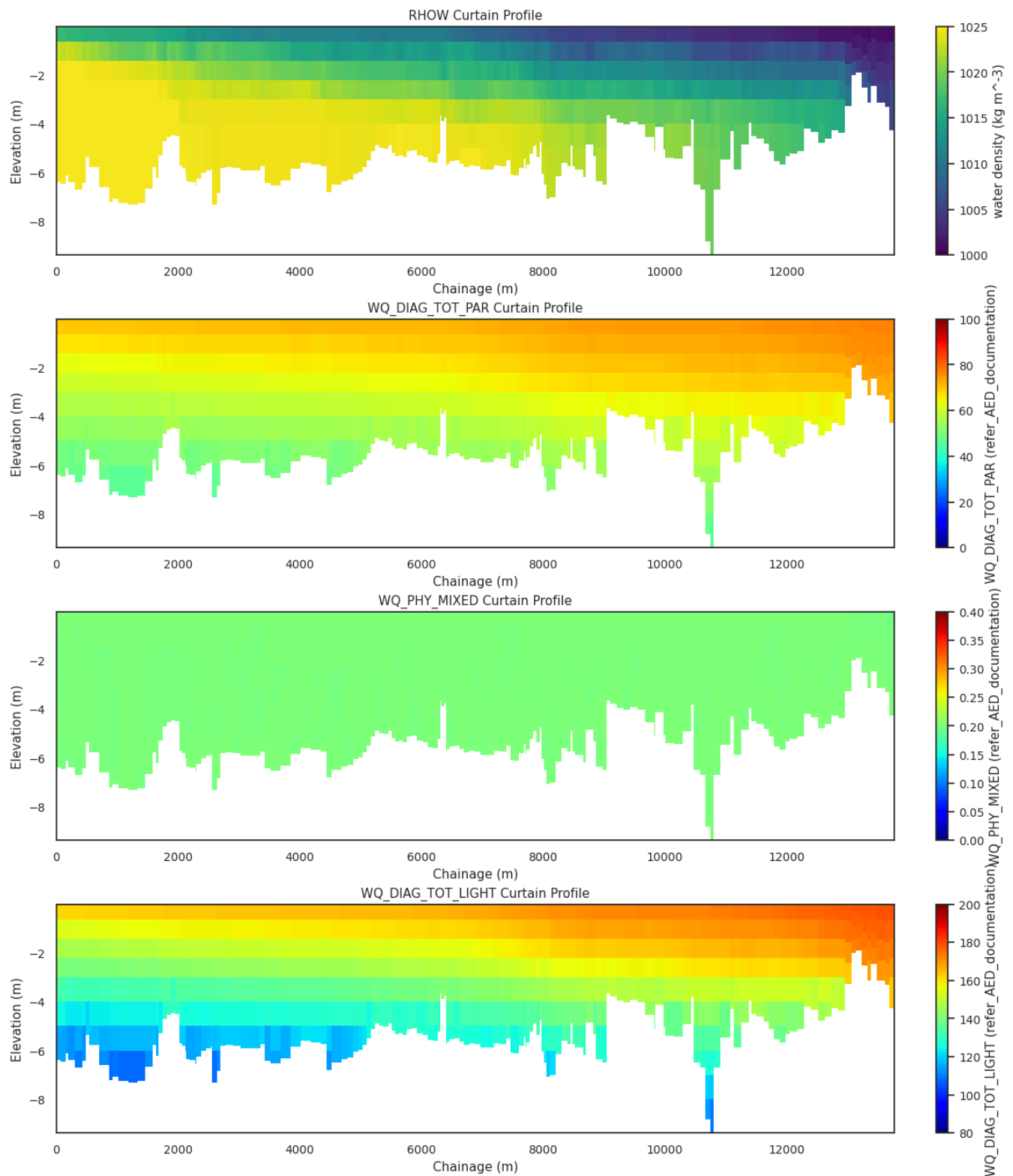


## Multi-variable curtain stack — light, mixing and density

Stack curtains for variables that together describe the **light-mixing-density** environment driving primary production:

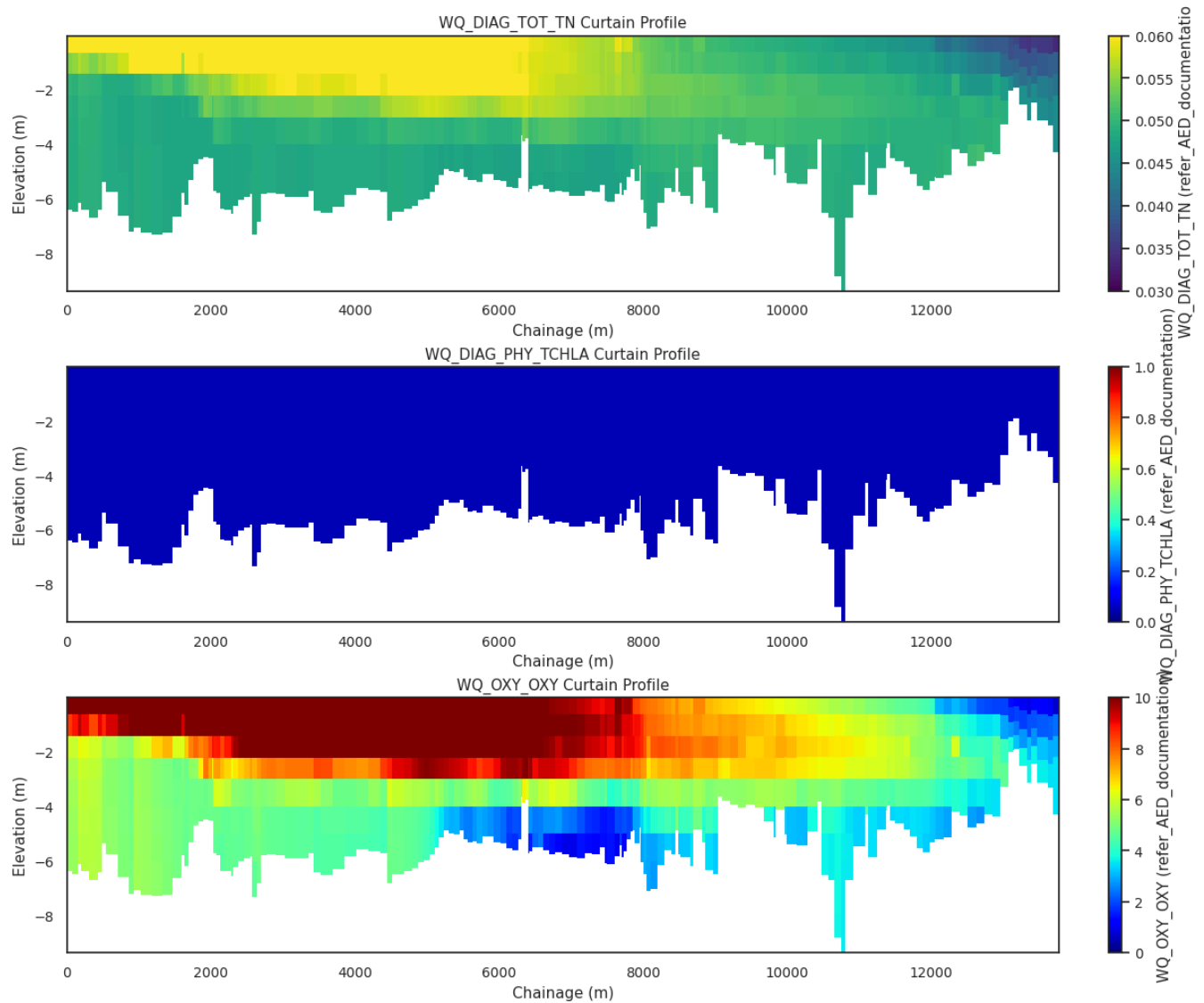
- `RHOW` — water density (top-down stratification).
- `WQ_DIAG_TOT_PAR` — photosynthetically active radiation.
- `WQ_PHY_MIXED` — mixed-phytoplankton biomass.
- `WQ_DIAG_TOT_LIGHT` — total light climate (depth-integrated).

Useful for visualising why phytoplankton biomass accumulates in some parts of the water column and not others.



## Nutrient / chlorophyll-a / oxygen curtains

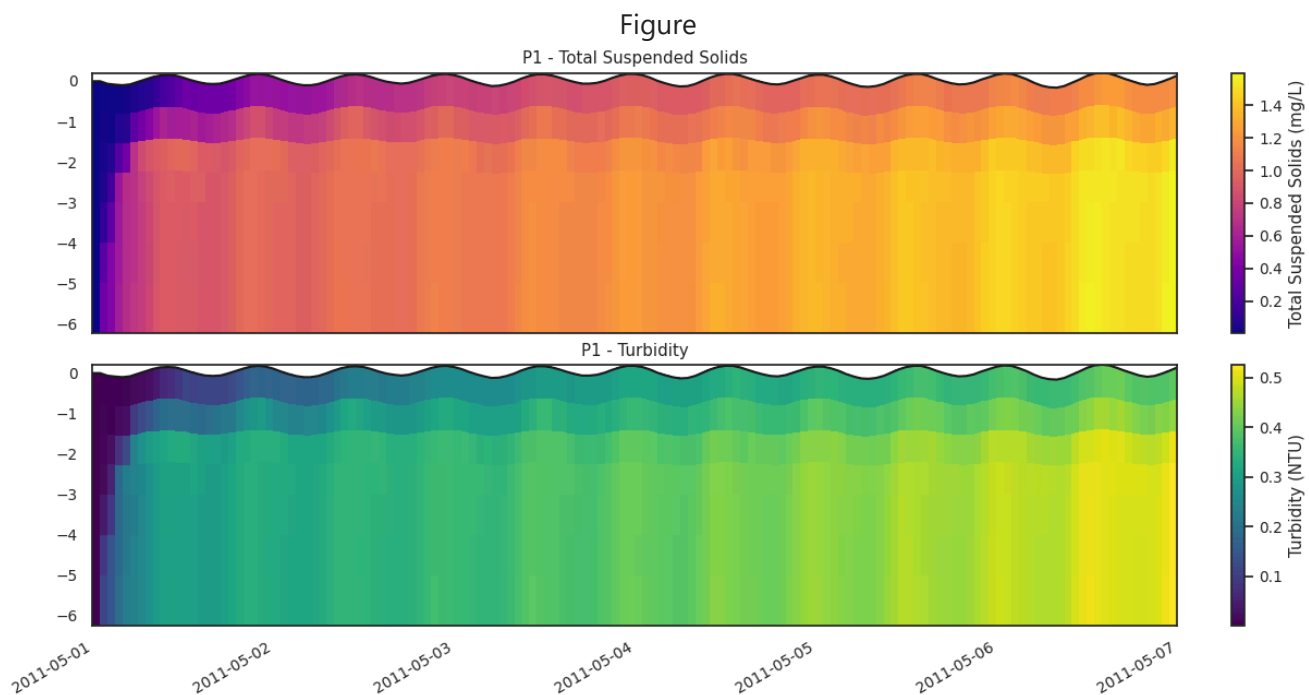
A second stack focused on biogeochemistry — total nitrogen ( `WQ_DIAG_TOT_TN` ), chlorophyll-a ( `WQ_DIAG_PHY_TCHLA` ), and dissolved oxygen ( `WQ_OXY_OXY` ). Useful for tracking nutrient → production → respiration pathways along the estuary.



## 8. Hovmoller diagrams (depth vs. time)

A Hovmoller diagram shows how a variable evolves through the **water column** at a single station over time — the y-axis is depth and the x-axis is time. It is the best single picture for spotting settling/resuspension cycles, productivity blooms or stratification events in a WQ field.

We re-define `locs` to a single high-precision station ( `P1` ) before plotting a combined **TSS + Turbidity** Hovmoller on a shared time axis.



## 9. Animation export (MP4)

Drive `matplotlib.animation.FuncAnimation` by calling the `tfv` plot patch's `set_time_current(i)` each frame. Here we animate the **Turbidity** field ( `WQ_DIAG_TOT_TURBIDITY` ) with velocity vector overlays. The output is written to `./outputs/Turbidity_Animation.mp4` .

**Requires** `ffmpeg` on `PATH` . If you don't have it, swap `FFmpegWriter()` for `animation.PillowWriter()` and change the suffix to `.gif` .

Ignoring fixed x limits to fulfill fixed data aspect with adjustable data limits.  
Saved animation to `outputs/Turbidity_Animation.mp4`

## 10. Interactive views (Jupyter widgets)

Switching the backend to `%matplotlib widget` enables interactive controls. `fv.plot_interactive` adds a time slider so you can scrub through the simulation without re-rendering — here on the depth-averaged **Turbidity** field with velocity vectors. The final cell closes any leftover figures and restores the inline backend so the rest of the notebook renders normally on GitHub.

**Note:** Interactive widgets only work in a **live** Jupyter session (with `ipymp1` installed) — they will appear as static placeholders when the notebook is rendered on GitHub. Leave the cell run once so the static fallback image is visible.

```
GridspecLayout(children=(HBox(children=(Play(value=0, description='Play', interval=500, layout=Layout(height='...
t=Layout(height='...
```

Ignoring fixed x limits to fulfill fixed data aspect with adjustable data limits.

Figure

